P2 Chapter 9: Differentiation

The Quotient Rule

The Quotient Rule

Just as we use the 'product rule' to differentiate a 'product', we use the '**quotient rule**' to differentiate a 'quotient' (i.e. division).

$$y = \frac{u}{v} = uv^{-1}$$

$$\frac{dy}{dx} = -uv^{-2}\frac{dv}{dx} + \frac{du}{dx}v^{-1}$$

$$= v^{-2}\left(v\frac{du}{dx} - u\frac{dv}{dx}\right)$$

$$= \frac{v\frac{du}{dx} - u\frac{dv}{dx}}{v^2}$$

✏ The quotient rule:

If $y = \frac{u}{v}$ then $\frac{dy}{dx} = \frac{v\frac{du}{dx} - u\frac{dv}{dx}}{v^2}$

Memorisation Tips: "Bottoms first!" The denominator (v) is the first term seen in the new denominator and numerator. The denominator gets squared. Note that in the numerator, we have $-$ instead of the $+$ seen in the Product Rule.

Examples

[Textbook] If $y = \frac{x}{2x+5}$, find $\frac{dy}{dx}$

?

[Textbook] Find the stationary point of $y = \frac{\sin x}{e^{2x}}$, $0 < x < \pi$

?

Examples

[Textbook] If $y = \frac{x}{2x+5}$, find $\frac{dy}{dx}$

$$\boldsymbol{u = x \qquad v = 2x + 5}$$
$$\boldsymbol{\frac{du}{dx} = 1 \qquad \frac{dv}{dx} = 2}$$
$$\boldsymbol{\frac{dy}{dx} = \frac{1(2x+5) - 2x}{(2x+5)^2} = \frac{5}{(2x+5)^2}}$$

Layout just like you would with the Product Rule.

[Textbook] Find the stationary point of $y = \frac{\sin x}{e^{2x}}$, $0 < x < \pi$

$$u = \sin x, \qquad v = e^{2x}$$
$$\frac{du}{dx} = \cos x, \qquad \frac{dv}{dx} = 2e^{2x}$$
$$\frac{dy}{dx} = \frac{e^{2x}\cos x - 2\sin x\ (e^{2x})}{e^{4x}} = 0$$
$$e^{2x}\cos x - 2\sin x\,(e^{2x}) = 0$$
$$e^{2x}(\cos x - 2\sin x) = 0$$
$$e^{2x} \neq 0 \quad \therefore \cos x = 2\sin x$$
$$\tan x = \frac{1}{2} \quad \rightarrow \quad x = 0.464\ (3sf)$$
$$y = \frac{\sin(0.464)}{e^{2\times 0.464}} = 0.177\ (3sf)$$

Whenever you have "fraction = 0", you can just multiply through by denominator.

As per advice earlier this chapter, try to factorise whenever you have 0 on one side.

As per trig chapter, divide by $\cos x$ whenever you have a mix of $\sin x$ and $\cos x$.

Test Your Understanding

Edexcel C3 Jan 2012 Q1a

Differentiate with respect to x, giving your answer in its simplest form,

(*b*) $\dfrac{\sin 4x}{x^3}$. **(5)**

?

Edexcel C3 June 2012 Q3

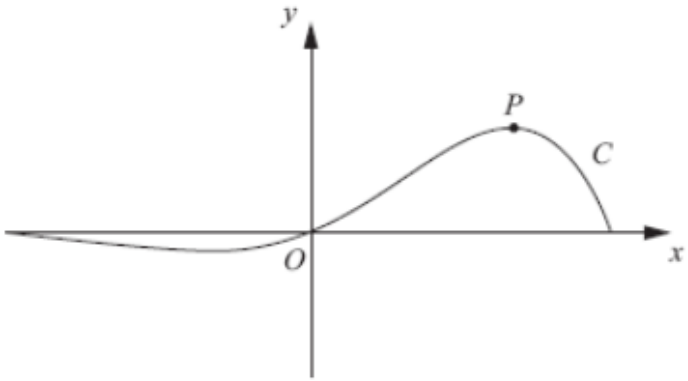


Figure 1

Figure 1 shows a sketch of the curve C which has equation

$$y = e^{x\sqrt{3}} \sin 3x, \quad -\frac{\pi}{3} \le x \le \frac{\pi}{3}.$$

(*a*) Find the x-coordinate of the turning point P on C, for which $x > 0$. Give your answer as a multiple of π. **(6)**

(*b*) Find an equation of the normal to C at the point where $x = 0$. **(3)**

?

Test Your Understanding

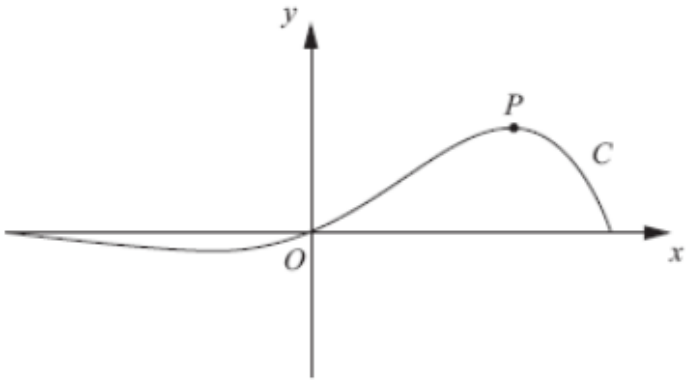
Edexcel C3 Jan 2012 Q1a

Differentiate with respect to x, giving your answer in its simplest form,

(*b*) $\dfrac{\sin 4x}{x^3}$. **(5)**

$$\frac{4x\cos 4x - 3\sin 4x}{x^4}$$

Edexcel C3 June 2012 Q3



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(a)	$\frac{dy}{dx} = \sqrt{3}e^{x\sqrt{3}}\sin 3x + 3e^{x\sqrt{3}}\cos 3x$	M1A1
	$\frac{dy}{dx} = 0 \qquad e^{x\sqrt{3}}(\sqrt{3}\sin 3x + 3\cos 3x) = 0$	M1
	$\tan 3x = -\sqrt{3}$	A1
	$3x = \frac{2\pi}{3} \Rightarrow x = \frac{2\pi}{9}$	M1A1
		(6)
(b)	At $x = 0$ $\quad \frac{dy}{dx} = 3$	B1
	Equation of normal is $-\frac{1}{3} = \frac{y-0}{x-0}$ or any equivalent $y = -\frac{1}{3}x$	M1A1
		(3)
		(9 marks)

Exercise 9.5

Pearson Pure Year 2
Page 73

Extension

[STEP I 2010 Q2] The curve $y = \left(\frac{x-a}{x-b}\right) e^x$, where a and b are constants, has two stationary points. Show that

$$a - b < 0 \text{ or } a - b > 4$$

(i) Show that, in the case $a = 0$ and $b = \frac{1}{2}$, there is one stationary point on either side of the curve's vertical asymptote, and sketch the curve.

(ii) Sketch the curve in the case $a = \frac{9}{2}$ and $b = 0$.

Homework Exercise

1 Differentiate:

a $\dfrac{5x}{x+1}$ **b** $\dfrac{2x}{3x-2}$ **c** $\dfrac{x+3}{2x+1}$ **d** $\dfrac{3x^2}{(2x-1)^2}$ **e** $\dfrac{6x}{(5x+3)^{\frac{1}{2}}}$

2 Differentiate:

a $\dfrac{e^{4x}}{\cos x}$ **b** $\dfrac{\ln x}{x+1}$ **c** $\dfrac{e^{-2x}+e^{2x}}{\ln x}$ **d** $\dfrac{(e^x+3)^3}{\cos x}$ **e** $\dfrac{\sin^2 x}{\ln x}$

3 Find the value of $\dfrac{dy}{dx}$ at the point $\left(1, \frac{1}{4}\right)$ on the curve with equation $y = \dfrac{x}{3x+1}$

4 Find the value of $\dfrac{dy}{dx}$ at the point (12, 3) on the curve with equation $y = \dfrac{x+3}{(2x+1)^{\frac{1}{2}}}$

5 Find the stationary point(s) of the curve C with equation $y = \dfrac{e^{2x+3}}{x}$, $x \neq 0$.

6 Find the equation of the tangent to the curve $y = \dfrac{e^{\frac{1}{3}x}}{x}$ at the point $\left(3, \frac{1}{3}e\right)$. **(7 marks)**

Homework Exercise

7 Find the exact value of $\frac{dy}{dx}$ at the point $x = \frac{\pi}{9}$ on the curve with equation $y = \frac{\ln x}{\sin 3x}$

8 The curve C has equation $x = \frac{e^y}{3 + 2y}$

a Find the coordinates of the point P where the curve cuts the x-axis. **(1 mark)**

b Find an equation of the normal to the curve at P, giving your answer in the form $y = mx + c$, where m and c are constants to be found. **(6 marks)**

9 Differentiate $\frac{x^4}{\cos 3x}$ with respect to x. **(4 marks)**

10 A curve C has equation $y = \frac{e^{2x}}{(x-2)^2}$, $x \neq 2$.

a Show that

$$\frac{dy}{dx} = \frac{Ae^{2x}(Bx - C)}{(x-2)^3}$$

where A, B and C are integers to be found. **(4 marks)**

b Find the equation of the tangent of C at the point $x = 1$. **(3 marks)**

Homework Exercise

11 Given that

$$f(x) = \frac{2x}{x+5} + \frac{6x}{x^2+7x+10}, \; x > 0$$

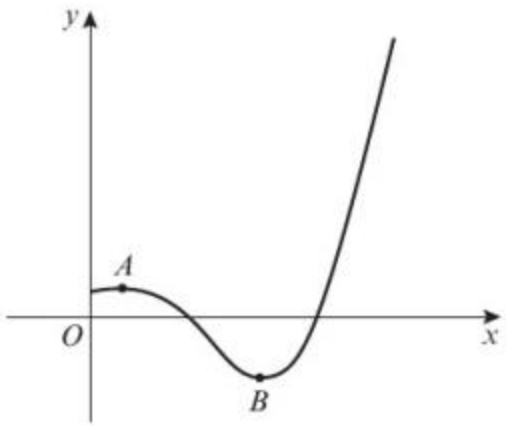
a show that $f(x) = \dfrac{2x}{x+2}$ **(4 marks)**

b Hence find $f'(3)$. **(3 marks)**

12 The diagram shows a sketch of the curve with equation $y = f(x)$, where

$$f(x) = \frac{2\cos 2x}{e^{2-x}}, \; 0 < x < \pi$$

The curve has a maximum turning point at A and a minimum turning point and B as shown in the diagram.



a Show that the x-coordinates of point A and point B are solutions to the equation $\tan 2x = \frac{1}{2}$ **(4 marks)**

b Find the range of $f(x)$. **(2 marks)**

Homework Answers

1 **a** $\dfrac{5}{(x+1)^2}$ **b** $-\dfrac{4}{(3x-2)^2}$ **c** $-\dfrac{5}{(2x+1)^2}$

d $-\dfrac{6x}{(2x-1)^3}$ **e** $\dfrac{15x+18}{(5x+3)^{\frac{1}{2}}}$

2 **a** $\dfrac{e^{4x}(\sin x + 4\cos x)}{\cos^2 x}$ **b** $\dfrac{1}{x(x+1)} - \dfrac{\ln x}{(x+1)^2}$

c $\dfrac{e^{-2x}(2x(e^{4x}-1)\ln x - e^{4x} - 1)}{x(\ln x)^2}$

d $\dfrac{(e^x+3)^2((e^x+3)\sin x + 3e^x\cos x)}{\cos^2 x}$

e $\dfrac{2\sin x\cos x}{\ln x} - \dfrac{\sin^2 x}{x(\ln x)^2}$

3 $\dfrac{1}{16}$ **4** $\dfrac{2}{25}$ **5** $(0.5, 2e^4)$

6 $y = \frac{1}{3}e$ **7** $\dfrac{6\sqrt{3} - 2\pi\ln\left(\frac{\pi}{9}\right)}{\pi}$

8 **a** $\left(\frac{1}{3}, 0\right)$ **b** $y = -\frac{1}{9}x + \frac{1}{27}$

9 $\dfrac{x^3(3x\sin 3x + 4\cos 3x)}{\cos^2 3x}$

10 **a** $\dfrac{(x-2)^2(2e^{2x}) - e^{2x}[2(x-2)]}{(x-2)^4} = \dfrac{2(x-2)^2e^{2x} - 2e^{2x}(x-2)}{(x-2)^4}$

$= \dfrac{2(x-2)e^{2x} - 2e^{2x}}{(x-2)^3} = \dfrac{2e^{2x}(x-2-1)}{(x-2)^3} = \dfrac{2e^{2x}(x-3)}{(x-2)^3}$

$A = 2, B = 1, C = 3$

b $y = 4e^2x - 3e^2$

11 **a** $\dfrac{2x}{x+5} + \dfrac{6x}{(x+5)(x+2)} = \dfrac{2x(x+2)}{(x+5)(x+2)} + \dfrac{6x}{(x+5)(x+2)}$

$= \dfrac{2x(x+2+3)}{(x+5)(x+2)} = \dfrac{2x(x+5)}{(x+5)(x+2)} = \dfrac{2x}{(x+2)}$

b $\dfrac{4}{25}$

12 **a** $f'(x) = -2e^{x-2}(2\sin 2x - \cos 2x) = 0$

$2\sin 2x - \cos 2x = 0 \Rightarrow \tan 2x = \frac{1}{2}$

b $-1.47 < f(x) < 6.26$